APPENDIX D

SOFTWARE TESTING DOCUMENTATION (STD)

RESQ SISWA UPSI VOLUNTEER PORTAL (WEB-BASED)

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This document is based on IEEE Std 1016-1998,
IEEE Standard for Software Design Description

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1.0 INTRODUCTION

The Software Testing Design Document offers a detailed insight into the testing strategy, objectives, and scope tailored for the ResQ Siswa UPSI Volunteer Portal.

1.1 Purpose

The purpose of this document is to test the functionality of ResQ Siswa UPSI Volunteer Portal. All testing deliverables and results in this document are related to ResQ Siswa UPSI Volunteer Portal..

1.2 Scope

The usability and functionality testing of ResQ Siswa UPSI Volunteer Portal are done incrementally through the project development process.

1.3 Definitions and Acronyms

Term	Definition
ResQ Siswa UPSI Volunteer Portal	Name of website developed
Student	A person who make an application to be a event volunteer
Organization	A person who accepts or declines the applications that were made by the students.
Application	A term refers to sign up to join something
TS	Test Scenario ID
TC	Test Case ID
R	Risk ID
IEEE	The Institute of Electrical and Electronics Engineers (IEEE) is a professional association

	headquartered in New York City that is dedicated to advancing technological innovation and excellence.
SRS	Software Requirements Specification- A document that completely describes all the functions of a proposed system and the constraints under which it must operate. For example, this document.
SDD	Software Design Document- A document that explains how a software product or a feature will be built to meet a set of technical requirements.
STD	Software Test Document- A document that involves artefacts that should be developed before and during the software or system testing.

1.4 References

I. Song, W. (n.d.). *Analysis of New Library Borrowing and Returning Mode Based on Book Transfer System*. https://doi.org/10.25236/iwass.2018.018

II. *Testing Object-oriented Systems*. (2024). Google Books. https://books.google.com.my/books?hl=en&lr=&id=P3UkDhLHP4YC&oi=fnd&pg=PR25&dq=Software+Testing+Design+for+Function+return+book+in+Library+Management+System&ots=_Idl3n2WE2&sig=OZI1-2bkHd9LoY0uAqihXP4WseY&redir_esc=y#v=onepage&q&f=false

III. IEEE. IEEE Std 829-1983 IEEE Standard for Software Test Documentation.

IV. IEEE. IEEE Std 829-1998 IEEE Standard for Software Test Documentation.

V. IEEE. IEEE Std 829-2008 IEEE Standard for Software and System Test Documentation.

1.5 Overview

The next chapter will discuss the details of the test plan of the system testing for ResQ Siswa UPSI Volunteer Portal. Then, the test scenarios and test cases will be explained in detail in the following chapter. The last chapter will show the test result of functional testing.

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2.0 TEST PLAN

2.1 Introduction

2.1.1 Purposes

This test plan for ResQ Siswa UPSI Volunteer Portal support the following objectives:

i. To detail the methods and activities required to prepare and conduct the test.

ii. To define the test environment needed to conduct the test.

2.1.2 Scope

This test plan only covers functionality tests of ResQ Siswa UPSI Volunteer Portal . This will focus on meeting the requirement of the app system and ensure the apps system is fit for purpose.

2.2 Test Item

All items which make up the ResQ Siswa UPSI Volunteer Portal will be tested. The test items include all system features in ResQ Siswa UPSI Volunteer Portal. The following documents are the basis of

items to be tested.

I. ResQ Siswa UPSI Volunteer Portal Project Backlog

II. ResQ Siswa UPSI Volunteer Portal Software Requirements Specification, version 3.0

III. ResQ Siswa UPSI Volunteer Portal Software Design Document, version 4.0

2.3 Features To Be Tested

Following table describes the features that will be tested based on the ResQ Siswa UPSI Volunteer Portal Software Requirements Specification (SRS).

Table 2.1: Features To Be Tested

Function ID	Description
F001	Register Account
F002	Log in
F003	Manage Profile
F004	Search Event
F005	View Event
F006	Sign up Event
F007	View Application
F008	Confirmation Application
F009	Add Events
F010	Edit Events
F011	Delete Events
F012	Manage Students
F013	View Student Application
F014	Confirmation Application
F015	Send Messages

2.4 Features Not To Be Tested

Aside from the functionality listed in Table 2.1, other aspects of the system will not be included in the system tests because they are not the main features that will be used in the system.

2.5 Approach

The test on ResQ Siswa UPSI Volunteer Portal is focused on functionality of the system. The following are the types of testing that will be used to test the system.

i. Functional Testing

ii. Usability Testing

For functional testing, test-driven development (TDD) testing method is used to test the initial function code of ResQ Siswa UPSI Volunteer Portal in which focus on the unit test. The unit test focuses on every piece of functions of the system. It will be done together with the coding process. Then, the test scenarios and test cases for each functionality are designed and created based on the system functional requirements. The functional testing process will be executed in each sprint after the coding process of each functionality to satisfy the Definition of Done (DoD). The usability testing will be performed only once after all the sprints have passed to evaluate the usability of the system. The usability testing will use the survey questionnaire method to collect the feedback and information from target users which is 3 members of the club and 7 students (non-member of the club).

2.6 Item Pass/Fail Criteria

The system must satisfy the following criteria to pass:

i. 100% test cases are executed

ii. 95% pass rate of test cases

iii. No serious defects

iv. No deadlock during the operation

2.7 Test Deliverables

The following deliverables will be generated, delivered, and combined in Software Test Document (STD) after test completion.

i. Test Plan

ii. Test Scenario and Test Case

iii. Test Result

2.8 Testing Tasks

The following table shows the set of tasks and activities of testing for ResQ Siswa UPSI Volunteer Portal. It identifies dependencies of other tasks and important constraints, such as test item availability, test resource availability, and deadlines.

Table 2.2: Testing Tasks

No.	Task	Target Date
1	Prepare Test Plan	22/10/24
2	Prepare Test Scenarios and Test Cases	26/11/24
3	Setup Test Environment	30/12/24
4	Perform Functional Test	01/01/25
5	Record and Analyse Test Result	04/01/25

2.9 Environmental Needs

ResQ Siswa UPSI Volunteer Portal will be hosted and tested at the developer site. The following shows the hardware and software requirements for ResQ Siswa UPSI Volunteer Portal that will be used in the test environment.

2.9.1 Hardware

Table 2.3: ResQ Siswa UPSI Volunteer Portal Hardware Requirements for Testing

Hardware Components	Requirements
Hardware Name	HP 15s-fq2717TU (Example Model)
Manufacturer	HP Inc.
Processor	Intel® Core™ i3-1115G4 @ 3.0GHz, 2 Cores, 4 Threads
Memory (RAM)	8.0 GB
Resolution	1920 x 1080 (Full HD)
Hard Disk	256 GB SSD (200 GB free)
Internet Connection	Unifi 100 Mbps

2.9.2 Software

Following table is the software requirements that will be used to testResq Siswa UPSI Volunteer Portal.

Table 2.4: ResQ Siswa UPSI Volunteer Portal Software Requirements for Testing

Software Components	Requirements
System Name	ResQ Siswa UPSI Volunteer Portal
Version	2.0
Internet Browser	Google Chrome
Localhost Server	Google Firestore/Github
Tools	VScode

2.9.3 Publications

The following documents are required to support the system testing:

i. Resq Siswa UPSI Volunteer Portal Project Backlog

ii. Resq Siswa UPSI Volunteer Portal Software Requirements Specification, version 3.0

iii. Resq Siswa UPSI Volunteer Portal Software Design Document, version 4.0

iv. Resq Siswa UPSI Volunteer Portal Test Plan

2.10 Responsibilities

Responsibilities may differ based on the actual statement of work. Since this project must be done individually, all the work will be done by the same person.

2.11 Schedule

The testing schedule of ResQ Siswa UPSI Volunteer Portal is the same as the target date in Table 2.2. The portal testing process will follow the iterative method and executed in parallel with the development.

2.12 Risks and Contingencies

This section identifies the risk issues that may impact the ResQ Siswa UPSI Volunteer Portal testing activities. The following table is the possible risks of testing with its mitigation plan.

Table 2.5: Risks and Mitigation Plan

Risk ID	Risk	Probability	Influence	Mitigation Plan
R1	Incorrect or incomplete stated requirements	High	High	Frequent demonstrations of new functionality
R2	Change in requirements during development process	High	High	Determine and fix user stories in project backlog
R3	Unavailability of servers or test environment	Medium	High	Get further details on the server or test environment at the developer side as as fast as possible.
R4	Strict time limits affect the testing flow	High	High	Follow the development schedule and timely notification of potential changes in the schedule.

3.0 TEST SCENARIO AND TEST CASES

3.1 Test Scenario for Functional Testing

Table 3.1: Test Scenario for Functional Testing

Test Scenario ID	Function	Priority
TS001	Register	High
TS002	Log in	High
TS003	Manage Profile	High
TS004	Search Event	Medium
TS005	View Event	High
TS006	Sign up Event	High
TS007	View Application	High
TS008	Confirmation Application	High
TS009	Add Events	High
TS010	Edit Events	High
TS011	Delete Events	High
TS012	Manage Students	High
TS013	View Student Application	Medium
TS014	Confirmation Application	High

3.1.1 TS001: Test Scenario of Register Account

Table 3.2: Test Scenario for register account

Test Scenario ID		TS001	
Test Function		Register Account	
Test Priority		High	
Description		Check the Functionality of Register Account	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Account registration is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC001	Register New Account in ResQ Siswa UPSI Volunteer Portal	Student Full Name: Sofiyya binti Basri Email: sofiyyabasri82@gmail.com Password: sof@8210 Phone number: 017-4356776 Organization Full Name: Ahmad Sufian Email: ahmadsufian88@gamil.com Password: sufian@2001 Phone Number: 012-4563211	ResQ Siswa Volunteer Portal system displays "Your account has been created successfully" and redirect to login page.
TC002	Register account with existing	Email: d097796@student.edu	ResQ Siswa UPSI Volunteer Portal

	email	-for student Email: sufian34@admin.com - for organization	system displays "This email is already in used".
TC003	Register account with entering alphabet in password field	Password: abc@student - for student Password abc@organization - for organization	ResQ Siswa UPSI Volunteer Portal system displays "Password should contain uppercase, number and special characters!"

3.1.2 TS002: Test Scenario for Login

Table 3.3: Test Scenario of login

Test Scenario ID		TS002	
Test Function		Login	
Test Priority		High	
Description		Check the Functionality of system login	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Successful log in to the e-Appointment app system.	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC004	Log in with valid account	Email: sofiyyabasri82@gmail.com Password: Sofiyya@2010 - for student	ResQ Siswa UPSI Volunteer Portal system displays "Logged in" and redirect to the main page.

		Email: ResqClub@gmail.com Password: ResQ@2013 - for organization	
TC005	Login with invalid email	Email: 1@student.edu -for student Email: 2@admin.com - for organization	ResQ Siswa UPSI Volunteer Portal system displays “Incorrect email. Please try again”.
TC006	Log in with invalid password	Password: @student - for student Password @organization - for organization	ResQ Siswa UPSI Volunteer Portal system displays “Incorrect password. Please try again”.
TC007	Log in with blank input	-	ResQ Siswa UPSI Volunteer Portal system displays “Please fill out this field”.

3.1.3 TS003: Test Scenario for Manage Profile

Table 3.4: Test scenario of manage profile

Test Scenario ID	TS003
Test Function	Manage Profile
Test Priority	High
Description	Check the Functionality of Manage Profile
Pre-condition	ResQ Siswa UPSI Volunteer Portal is

		completely set up in test environment	
Post-Condition		Successful to manage profile	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC008	Edit name	Student Full Name: Sofiyya binti Basri	Updated name should be displayed.
TC009	Edit profile picture	Upload new image of student and organization: profile.jpg	New profile picture should be displayed.
TC010	Edit Password	New Password: Syifaa@2001	Passwords should be updated successfully.

3.1.4 TS004: Test Scenario for Search Events

Table 3.5: Test scenario of search events

Test Scenario ID		TS004	
Test Function		Search Events	
Test Priority		Medium	
Description		Check the Functionality of search available events	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Available events list displayed	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC011	Click “Event” button	-	ResQ Siswa UPSI Volunteer Portal

			system redirects to event page.
TC012	Enter keywords of appointment and click "Search" icon	Search: "*Gotong-royong Perdana*", "*Bantuan Pasca-Banjir*".	ResQ Siswa UPSI Volunteer Portal system displays the available event.
TC013	Click "Search" icon without entering any keywords	-	ResQ Siswa UPSI Volunteer Portal system remains at the events page.

3.1.5 TS005: Test Scenario for View Event

Table 3.6: Test scenario of view event

Test Scenario ID		TS005	
Test Function		View event	
Test Priority		High	
Description		Check the Functionality of view details of available events	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Details of event is displayed on screen	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC014	Click "Event" button	-	Event list displayed on screen.

3.1.6 TS006: Test Scenario for Sign Up Events

Table 3.7: Test scenario of sign up events

Test Scenario ID		TS006	
Test Function		Sign up events	
Test Priority		High	
Description		Check the Functionality of sign up events	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Sign up events is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC015	Click “Sign up” button	-	ResQ Siswa Volunteer Portal system displays events details
TC016	Click “Sign up” button	-	ResQ Siswa UPSI Volunteer Portal system displays information to be filled
TC017	Fill in the blanks with the correct information	Name: Sofiyya binti Basri IC: 0102070984 Email: sofiyya82@gmail.com	ResQ Siswa UPSI Volunteer Portal system displays “Successful sign up the events”

3.1.7 TS007: Test Scenario for View Application

Table 3.8: Test scenario of view application

Test Scenario ID		TS007	
Test Function		View Application	
Test Priority		High	
Description		Check the Functionality of View Application by students	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		View Application is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC018	Click “My applications” button		List of applications displayed on screen.

3.1.8 TS008: Test Scenario for Confirmation Application

Table 3.9: Test scenario of confirmation application

Test Scenario ID	TS008
Test Function	Confirmation Application
Test Priority	High
Description	Check the Functionality of Confirmation Application by students
Pre-condition	ResQ Siswa UPSI Volunteer Portal is completely set up in test environment

Post-Condition		Confirmation Application is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC019	Click “Confirm” button	-	ResQ Siswa Volunteer Portal system displays “Your Confirmation is successfully”
TC020	Click “Reject” button		ResQ Siswa UPSI Volunteer Portal system displays “Are you sure you want to reject?”.

3.1.9 TS009: Test Scenario for Add Events

Table 3.10: Test scenario of add events

Test Scenario ID		TS009	
Test Function		Add events	
Test Priority		High	
Description		Check the Functionality of Add Events by organization	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Add event is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC021	Click “add” button	-	An event detail form displayed on

			screen.
TC022	Add new event with fill up all the details of title, date, time and place and click “Save” button	Title: “*Bantuan Pasca Banjir*” Description: To help people that have been affected by flood Date: 23/2/2024 Time: 8:00 am - 5:00 pm Place: Pasir Mas, Kelantan	ResQ Siswa UPSI Volunteer Portal system displays “Events added successfully”.
TC023	Add new appointment with blank input and click “Save” button	-	ResQ Siswa UPSI Volunteer Portal system displays “Please fill out this field”.

3.1.10 TS010: Test Scenario for Edit Events

Table 3.11: Test scenario of edit events

Test Scenario ID		TS010	
Test Function		Edit events	
Test Priority		High	
Description		Check the Functionality of Edit events by organizations	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Edit events is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC024	Click “event” button	-	Events lists displayed on screen

TC025	Click ‘edit’ button on specific event	-	Edit event details form displayed on screen.
TC026	Edit event details	Title: “*Bantuan Pasca Banjir*” Description: To help people that have been affected by flood Date: 23/2/2024 Time: 9:00 am - 6:00 pm Place: Terengganu	ResQ Siswa UPSI Volunteer Portal system displays “The event details updated successfully”

3.1.11 TS011: Test Scenario for Delete Events

Table 3.12: Test Scenario of delete events

Test Scenario ID		TS011	
Test Function		Delete Events	
Test Priority		High	
Description		Check the Functionality of Delete Events by organization	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Delete Events is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC027	Click “Delete” button on selected event	-	ResQ Siswa Volunteer Portal system displays “The appointment deleted successfully”. The new

			list of appointments displayed on screen.

3.1.12 TS012: Test Scenario for Manage Students

Table 3.13: Test Scenario of manage students

Test Scenario ID		TS012	
Test Function		Manage Students	
Test Priority		High	
Description		Check the Functionality of Manage Students by organization	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		Manage Students is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC028	Click “view student profile” button	-	ResQ Siswa Volunteer Portal system displays selected student profile
TC029	Click “Delete” button on selected student	-	ResQ Siswa Volunteer Portal system displays “The appointment deleted successfully”

3.1.13 TS013: Test Scenario for View Student Application

Table 3.14: Test scenario of view student application

Test Scenario ID		TS013	
Test Function		View Student Application	
Test Priority		Medium	
Description		Check the Functionality of view student application by organization	
Pre-condition		ResQ Siswa UPSI Volunteer Portal is completely set up in test environment	
Post-Condition		View Student Application is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC030	Click selected event button		ResQ Siswa Volunteer Portal system displays the details of the event including the total of applications

3.1.14 TS014: Test Scenario for Confirm Students Application

Table 3.15: Test scenario of confirm students application

Test Scenario ID	TS014
Test Function	Confirm Student Application
Test Priority	High
Description	Check the Functionality of Confirm Students Application by organization
Pre-condition	ResQ Siswa UPSI Volunteer Portal is

		completely set up in test environment	
Post-Condition		Confirm Student Application is successfully	
Test Case ID	**Test Case**	**Input**	**Expected Result**
TC031	Click “accept” button on selected student applications	-	ResQ Siswa Volunteer Portal system displays “Accept has been successfully” and will notify the students.
TC032	Click “reject” button on selected student applications	-	ResQ Siswa Volunteer Portal system displays “Reject has been successfully” and will notify the students.

4.0 TEST RESULT

4.1 Test Result for Functional Testing

Table 4.1: Test Result of Functional Testing

Test Scenario ID	Test Cases ID	Result	Status
TS001	TC001	PASS	DONE
	TC002	PASS	DONE
	TC003	PASS	DONE
TS002	TC004	PASS	DONE
	TC005	PASS	DONE
	TC006	PASS	DONE
	TC007	PASS	DONE
TS003	TC008	PASS	DONE
	TC009	PASS	DONE
	TC010	PASS	DONE
TS004	TC011	PASS	DONE
	TC012	PASS	DONE
	TC013	PASS	DONE
TS005	TC014	PASS	DONE
TS006	TC015	PASS	DONE
	TC016	PASS	DONE
	TC017	PASS	DONE
TS007	TC018	PASS	DONE

TS008	TC019	PASS	DONE
	TC020	PASS	DONE
TS009	TC021	PASS	DONE
	TC022	PASS	DONE
	TC023	PASS	DONE
TS010	TC024	PASS	DONE
	TC025	PASS	DONE
	TC026	PASS	DONE
TS011	TC027	PASS	DONE
TS012	TC028	PASS	DONE
	TC029	PASS	DONE
TS013	TC030	PASS	DONE
TS014	TC031	PASS	DONE
	TC032	PASS	DONE